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Papop Lekhapanyaporn

Experience

GNS Systems GmbH | HPC System Engineer — Jan 2026 — Present

- Building a high-performance in-house replacement for legacy cluster management software in **C++**.

Redcare Pharmacy | Working student — Aug 2023 — Jan 2026

- Built a **Python** monitoring tool that tracked Gematik e-prescription updates and sent automated event notifications, cutting manual tracking effort for the team.
- Built a local **Wiremock/Python** test environment for e-prescription validation; automated template-to-mock conversion to enable early compliance checks.
- Wrote **Python** scripts for bulk e-prescription refactoring — type conversions and field edits — accelerating test-case generation.

Education

RWTH Aachen University | B.Sc Computer Science — Oct 2022 — Sep 2025

- **Thesis**: Benchmarked automated memory management on **NVIDIA GH200**, characterizing performance trade-offs across diverse memory access patterns.

Projects

HPC Application | C++ OpenMP MPI — Oct 2024 — Jul 2025

- Parallelized sparse matrix-vector multiplication, merge sort, and K-means clustering in **C++** with **OpenMP** across a shared-memory cluster environment.
- Profiled an **MPI** heat conduction simulation with **Vampir** and **Score-P**, diagnosing and resolving load imbalance and communication overhead.

SPOS — Embedded OS on ATmega664 | C AVR — Apr 2024 — Jul 2024

- Built a bare-metal OS for the **ATmega664** in **C/AVR-GCC**, implementing interrupts, critical sections, preemptive task scheduling, and dynamic memory spanning internal SRAM and external **23LC1024** RAM.
- Demonstrated OS capabilities through LED matrix output, joystick input, and a real-time **Snake** game running on the custom scheduler.

Asclepius — ASL Translator | Python OpenCV TensorFlow — Mar 2022 — Jun 2022

- Led a team of five to build a real-time ASL-to-text translator using gesture recognition — **1st Place** at the **Microsoft APAC AI for Accessibility Hackathon 2022**.
- Designed and trained the recognition model in **TensorFlow** and **OpenCV**; coordinated team planning and delivery.

Blade Defect Detection (Philips) | Python OpenCV NumPy scikit-learn — May 2023

- Built a shaving blade defect detector in **Python**, **OpenCV**, and **NumPy** achieving **96% accuracy** — **2nd Place** at the **Fraunhofer Hackathon 2023**.
- Unified the team's code into a single pipeline and evaluated performance with **scikit-learn** confusion matrix metrics.

Food Busters — Food Recognition App | Python TensorFlow OpenCV — Sep 2023 — Dec 2023

- Trained a **TensorFlow** image classifier to identify food from user photos, powering diet suggestions and food-waste reduction features.
- Integrated the model into a full-stack web app with a cross-functional team, targeting sustainability and healthier eating habits.

Web Portfolio | Next.js Prisma.js Vercel — Jan 2023 — Present

- Built and deployed a personal portfolio in **Next.js** and **Prisma**, hosted on **Vercel**.

Secret santa | Next.js — Nov 2022 — Dec 2022

- Built and deployed a **Next.js** app that generates fair, loop-free Secret Santa pairings.

SPACE AC | Software Engineer — Oct 2020 — Mar 2022

- **SPOROS** | Arduino C Python Qt5 — Nov 2020 — Jul 2021
  - Led full-stack embedded development in **Arduino (C)** for two autorotating payloads and a CanSat relay, plus a **Python/Qt5** ground station with real-time telemetry visualization.

- Designed the inter-payload communication protocol enabling mid-air telemetry relay; secured **3$^{\text{rd}}$ place** at the **Annual CanSat Competition 2021**.

- **AlienSat** | Arduino C Python Qt5 — Aug 2021 — Feb 2022
  - Developed real-time firmware for a **thermal camera** CanSat payload, streaming temperature arrays for PM2.5–heat correlation analysis.
  - Built live thermal visualization in the **Python/Qt5** ground station and designed the end-to-end payload-to-ground communication protocol.
- **Passenger Balloon** | Arduino C Python — Oct 2020 — Mar 2022
  - Flew **three high-altitude balloon missions** deploying CubeSat payloads to **35 km**, using **Arduino** and **Raspberry Pi** for autonomous imaging and atmospheric sensing.
- **Mentoring** | — Oct 2021 — Mar 2022
  - Designed and ran a structured onboarding program covering programming fundamentals, project workflow, and the team's hardware/software stack.

Extracurricular Activities

Interact Club | President, District Vice President — May 2020 — Mar 2022

- Led a **Rotary International**-supported student club; organized annual events and drove cross-club community service initiatives.

Technical Skills

Language: Python, C/C++, Java, Typescript/Javascript, HTML, CSS/SCSS, Haskell, SQL, PostgreSQL, Sass

Framework: CMake, CUDA, OpenMP, MPI, AVR, Arduino, Pytorch, Tensorflow, Scikit-learn, Jupyter, Conda, Pandas, NumPy, React, Next.js, Node.js, Prisma.js

Tools: Git, Github/Gitlab, RaspberryPI, XCTU(Zigbee), Vercel, Linode, Digital Ocean

Language

Thai: Native **English**: C2 **German**: C1